

On the Agile Transformation at Norlys

Beltran Aceves¹

Abstract—Norlys, a Danish cooperative formed from over forty merged companies, is undergoing an agile transformation. Using Weinberg’s overstructured management theory as the primary analytical lens, this paper identifies which managerial fallacies Norlys rejects. The findings are then discussed in relation to Lyytinen, Rose, and Yoo’s hyper-learning framework (which explains what replaces the rejected patterns) and Mintzberg’s organization configurations (which expose remaining structural tensions). The post-merger diversified form has not stabilized, and the middle line still carries command-and-control reflexes. Behavioral change must be paired with structural realignment.

I. INTRODUCTION

Norlys, a Danish cooperative formed from over forty merged organizations, currently employs about 4,500 people serving 3.5 million customers. Its One 2027 initiative pushes agile methods across the firm [1] amid clashing cultures: Telia’s performance-driven approach meeting Norlys’s cooperative tradition [2].

Agile transformations fail most often due to managerial thinking, not the method itself. When leaders expect fixed steps, treat people as interchangeable parts, or paste the same solution onto every problem, the transformation stalls. Weinberg [3] called this **overstructured management**—imposing programmer logic onto human systems.

This paper asks: **How does Norlys avoid the overstructured thinking that derails similar initiatives, and what structural tensions remain?**

Weinberg’s overstructured management theory is chosen as the primary analytical lens because the case data—particularly the agile coach’s critique of managerial thinking—reveal a pattern of fallacious reasoning that Weinberg’s framework is uniquely equipped to diagnose. The analysis proceeds in two steps. First, Weinberg’s DEAL model and five fallacies are applied systematically to identify which fallacies Norlys rejects and how. Second, the findings are discussed in relation to Lyytinen, Rose, and Yoo’s [4] hyper-learning framework (which explains what Norlys builds in place of overstructured patterns) and Mintzberg’s [5] configurations (which explain why the transformation remains uneven). The empirical base is five semi-structured interviews with Norlys stakeholders conducted in spring 2026 [6].

II. THEORETICAL FRAMEWORK

A. Overstructured Management

Weinberg [3] proposed two diagnostic tools for managerial thinking. The DEAL model distinguishes successful managers from unsuccessful ones by where they attribute outcomes. **Successful** managers locate outcomes in **Effort** (what they did) and **Ability** (what they are capable of). **Unsuccessful** ones drift toward **Difficulty** (the task was hard) and **Luck** (circumstances betrayed them). The first pair implies control; the second implies helplessness.

The five management fallacies describe specific overstructured moves. **Sequence** assumes one fixed order of steps, followed faithfully, produces transformation. **Choice** flattens complex situations into binary forks. **Modularization** treats roles and people as swappable parts. **Iteration** assumes that repeating the same action yields progress. **Recursion** presses a single frame onto problems that demand several. Together, they describe the programmer’s mindset imported into management, where everything looks deterministic, decomposable and repeatable.

III. ANALYSIS: REJECTING OVERSTRUCTURED MANAGEMENT

Five Norlys stakeholders were interviewed for this study. Their roles and abbreviations (used in the table below) are: Agile Coach (AC) Niels Platz, Lead Developer (LD) Kasper Harry Munck, Scrum Master (SM) Eva-Maria Bücking-Rasmussen, Product Owner (PO) Birgitte Frendrup Jakobsen, and Enterprise Architect (EA) Rasmus Witt. The analysis draws primarily on Niels, who with seventeen years of practice and a CS background grounded in organizational theory is a sharp critic of the patterns Weinberg named.

A. The DEAL Model: Effort and Ability in Practice

Niels attributes outcomes to effort and ability: **“We started with behavior: trying to create a common agreement of what we are working on”** [7]. Setbacks are capability gaps: **“We cannot just give them free rein, because they do not know what to do with it yet”** [7]. Eva-Maria, the scrum master, calls her role a **“guardrail”**—a check on drift, not a controller. The negative case is the manager who removed all scrum masters from platform teams, treating the role as a **difficulty** (an obstacle) rather than an **ability** (a capability teams

¹Aceves is with Department of Computer Science, Aalborg University, Aalborg, Denmark, baceve21@student.aau.dk.

TABLE I. CONCEPTS MAPPED TO INTERVIEW EVIDENCE WITH COUNTS. ROLE ABBREVIATIONS: AC = AGILE COACH, LD = LEAD DEVELOPER, SM = SCRUM MASTER, PO = PRODUCT OWNER, EA = ENTERPRISE ARCHITECT. FULL TRACEABILITY WITH VERBATIM QUOTES IN APPENDIX.

Concept	Empirical evidence (interviews)	Count
Weinberg — DEAL: Effort/ Ability	AC frames setbacks as capability gaps; SM describes her role as a “guardrail” not a controller; LD attributes decisions to developer ability; EA narrates career as accumulated effort; PO judges trade-offs by gut feel (Section VI.A.a)	5
Weinberg — DEAL: Difficulty/ Luck	No current Norlys manager makes this attribution. The only instance is a removed third party described by AC (Section VI.A.b)	0
Weinberg — Anti-Sequence	AC frames transformation as evolution, calls TPM role an “antipattern but a stepping stone”; PO expects her own role to disappear in three years; LD benefits from stable QBR cycles; SM adapts practices; EA describes standards that evolve not dictate (Section VI.A.c)	5
Weinberg — Anti-Modularization	AC describes the anchor case: a manager fired all Scrum Masters thinking the role was a swappable function — teams lost direction, role reinstated, manager removed. PO reports TPM role adapts to each team (Section VI.A.e)	2
Weinberg — Anti-Choice	AC rejects bottom-up vs. top-down binary; EA confirms hybrid model; SM balances quality and speed; PO describes process deviation as a spectrum; LD uses confidence voting (Section VI.A.d)	5
Weinberg — Anti-Iteration	SM dropped sprint reviews when they stopped producing value; AC says “real world has a veto right”; PO reports past speed-over-quality choices are not repeated; LD avoids early lifecycle versions (Section VI.A.f)	4
Weinberg — Anti-Recursion	AC works “between the lines” crossing team and management boundaries; SM has a non-IT background that prevents single-frame thinking; EA career arc traces four distinct perspectives (Section VI.A.g)	3
Lyytinen et al. — Distributed gate-keeping	SM listens for unspoken obstacles; PO describes a prioritisation funnel; EA operationalises gate-keeping as friction; AC scans for practices; LD evaluates third-party APIs (Section VI.B.a)	5
Lyytinen et al. — Brokering external knowledge	EA describes post-Telia cultural integration; AC warns Telia’s model needed translation; LD brokers third-party API specs; PO (non-IT background) translates between business and engineering (Section VI.B.b)	4
Lyytinen et al. — Simple design rules	LD uses the “picture” heuristic; PO applies the 80/20 capacity rule; EA maintains platform standards; AC uses Q Planning lock as a boundary rule; SM uses T-shirt sizing (Section VI.B.c)	5
Lyytinen et al. — Peer networks	AC frames code review as bidirectional teaching; SM calendars monthly knowledge-sharing sessions; LD walks new services through the team; EA runs an architecture studio (Section VI.B.d)	4
Lyytinen et al. — Parallel ambidexterity	AC designed the IP sprint as dedicated exploration; SM deliberately “cheats” the IP sprint to explore immediately rather than waiting (Section VI.B.e)	2
Mintzberg — Middle line resistance	AC names middle management the hardest place to change; anchor case acts on machine logic; EA reports leadership turnover destabilises priorities; PO makes decisions then apologises; LD notes trust substitutes for supervision (Section VI.C.a)	5
Mintzberg — Technostructure pull	AC describes finance requiring story points for cost calculation; PO calls the funnel “a dogfight somewhere in leadership”; LD has platform teams standardising; EA designs friction; SM reports mandatory release windows (Section VI.C.c)	5

had not yet built). Teams lost direction, and the manager was removed [7].

B. The Sequence Fallacy: Evolution Over Revolution

Niels rejects a single sequence: **“You can take revolutions... or you can take evolutions... Evolutions have less risk”** [7]. SAFe influences appear but no fixed roadmap was adopted—Q Planning, the TPM role, and the feature factory are openly flagged as imperfect stepping stones. The TPM role is called **“an antipattern of huge proportions, but it is a stepping stone”** [7]. The ordering is deliberate: behavior first, structure second. **“You can start by doing an organizational trans-**

formation, which always messes up, because you start by just putting people in a new box, and they will act as they did before” [7].

C. The Modularization Fallacy: The Scrum Master Case

A new manager fired every scrum master on the platform teams, reasoning like a programmer: if a function sits outside the spec, deleting it should be free. Teams built a new tool, missed what customers wanted, and direction evaporated. The role was reinstated and the manager removed [7]. The scrum master was not a module but a catalyst for knowledge sharing,

customer connection and team self-organization—value that surfaced only after the role vanished.

D. The Choice Fallacy: Hybrid Coordination

Niels refuses the bottom-up versus top-down binary: **“you need to work on all different levels”** [7]. Rasmus confirms a hybrid model: **“Teams coordinate horizontally... while priorities are set top-down”** [2].

E. The Iteration and Recursion Fallacies

Niels catches himself on iteration: **“Agile is the baseline for this, and it always will be.”** His remedy is to test the claim: **“let the real world have a veto right”** [7]. The recursion fallacy meets a counterweight in the agile coach role, which works **“between the lines”**—crossing team, management and methodology boundaries—applying multiple frames where management sees one [7].

The table below summarizes the empirical support for each concept and points to the appendix for full traceability (including verbatim quotes, indicators, and the cross-cutting anchor case).

IV. DISCUSSION: RELATING FINDINGS TO HYPER-LEARNING AND ORGANIZATION THEORY

The Weinberg-based analysis showed what Norlys **refuses**: each of the five fallacies is rejected in practice. This section relates those findings to two other course theories. First, Lyytinen, Rose, and Yoo’s [4] hyper-learning framework is used to interpret what Norlys adopts in place of overstructured patterns. Second, Mintzberg’s [5] organization configurations provide a structural explanation for why the transformation remains uneven.

A. Hyper-Learning as the Replacement

Lyytinen, Rose, and Yoo describe how organizations learn through four parallel routines. These appear across interviews, though unevenly.

a) Distributed Gate-Keeping:

Multiple roles act as gate-keepers. The agile coach scans for new practices, the scrum master watches team dynamics and flags organizational barriers, and the enterprise architect tracks technology trends and turns them into guidelines. Rasmus: **“Guidelines exist for a reason... If you choose another approach, you must meet the same standards”** [2]. Gate-keeping is distributed; no single function owns it.

b) Brokering External Knowledge:

The post-merger context makes brokering primary. Norlys absorbed Telia Denmark, dragging a performance-driven culture into a cooperative frame [2]. Rasmus describes the friction: **“Cultures merge too, and it takes time... there were clashes initially.”** The organization translates between knowledge bases continuously.

c) Simple Design Rules:

Simple heuristics steer action without prescribing every move. Story points gauge relative size, not precise forecasts. The **“picture”** heuristic (an 8+ point story is too fuzzy) pushes decomposition: **“If something is estimated at eight, it is basically a picture”** [8]. T-shirt sizing and confidence votes coordinate fast, without bureaucratic overhead.

d) Peer Networks:

Code review and IP sprints (Innovation and Planning sprints, the sixth sprint of each quarter) enable horizontal knowledge sharing. Kasper describes code review as bidirectional learning: **“The inexperienced person learns, and the experienced person teaches”** [8]. IP sprints provide dedicated exploration alongside delivery [9].

e) Parallel Ambidexterity:

The signature hyper-learning pattern is exploration and exploitation coexisting in the same units. Teams ship in regular sprints and probe in IP sprints, with fast switching between modes. This matches Lyytinen’s parallel ambidexterity closely, even if Norlys never names it that way. The fit is imperfect: peer networks operate more formally than Lyytinen’s model, and Q Planning and the TPM role point to a more SAFe-shaped structure than a pure hyper-learner would tolerate.

B. A Structural Interpretation

The analysis above shows that Norlys rejects overstructured management, and the hyper-learning routines above show what it builds instead. Yet the transformation is uneven—some teams operate in mature learning routines while others stay dependent on command-and-control. Mintzberg’s [5] organization configurations explain why.

a) The Diversified Configuration:

Norlys’s post-merger shape fits the diversified configuration: a holding company over multiple divisions (energy, telecommunications, streaming), each carrying legacy structures. The coordinating mechanism is standardization of outputs (KPIs, financial targets). The One 2027 strategy counterpulls against balkanization toward integration.

b) The Push Toward Adhocracy:

The target is Mintzberg’s innovative form (adhocracy), where mutual adjustment runs among multidisciplinary teams coordinated by liaison devices (agile coaches, enterprise architects, scrum masters). The coordinating mechanism shifts from output standardization to mutual adjustment. Norlys sits in transition. The behavioral changes point toward adhocracy; the structure lags behind.

c) The Middle Line as Resistance Point:

Niels names it directly: **“Middle management is maybe the hardest place to change”** [7]. The reasons are structural. Middle managers’ authority rested on direct supervision, the coordinating mechanism of machine configurations. Agile swaps this for mutual adjustment, eroding their traditional

power base. The Scrum Master removal illustrates a manager acting on machine logic—simplify the structure, strip what is not essential—which is catastrophic during a transition toward adhocracy, where liaison devices carry the coordination load.

d) *The Pull of the Technostructure:*

Finance, compliance and analyst functions apply machine-configuration logic to an adhocracy-in-formation. When finance required story points on every closed task for cost calculation, it was standardizing work processes onto a structure that no longer fit [7]. The mandate is rational from a machine perspective (uniform data for cost analysis) but disrupts team-level estimation practices.

e) *What the Structural Lens Reveals:*

The behavioral rejection of overstructured management clusters in the operating core (teams, scrum masters, agile coaches) and the strategic apex (IT leadership, One 2027), while the middle line and technostructure retain overstructured assumptions. Unevenness does not signal failed implementation but incomplete structural transition.

V. CONCLUSION

Norlys avoids overstructured management by systematically rejecting Weinberg’s five fallacies. Its leaders anchor outcomes in effort and ability rather than difficulty and luck. They choose evolution over revolution, refuse to treat roles as interchangeable modules, resist the bottom-up versus top-down binary, and test whether iteration still serves rather than repeating agile mantras mindlessly. In place of overstructured patterns, Norlys builds hyper-learning routines: distributed gate-keeping, knowledge brokering, simple design rules, peer networks, and parallel ambidexterity.

Mintzberg’s lens exposes what remains. The transition from a diversified post-merger configuration toward adhocracy is behaviorally advanced but structurally incomplete. The middle line and technostructure still operate on assumptions from the machine configuration. Sustainable transformation requires redefining middle management roles and limiting the technostructure’s reach into team-level practices.

The study draws on interview data from a single organization mid-transformation. Transformation advocates (coaches, architects, scrum masters) speak louder here than skeptics or frontline developers. Future work could track how the structural transition unfolds over time, and whether Norlys’s hyper-learning routines hold up as its One 2027 program matures.

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VI. APPENDIX: TRACEABILITY FROM THEORETICAL CODES TO EMPIRICAL MATERIAL

This appendix documents how the empirical material—five semi-structured interviews with Norlys stakeholders—was used to map Norlys’s agile software development practice to the three theoretical lenses that structure the main paper: Weinberg’s overstructured management theory, Lyytinen, Rose, and Yoo’s hyper-learning framework, and Mintzberg’s organization configurations. The appendix supports the empirical traceability of the main paper’s argument by linking each theoretical code to interview indicators. Codes with stronger empirical support are documented with multiple indicators; codes with weaker support are documented with one representative indicator; absences are documented as such.

The five interviews are referred to throughout by interviewee role: **Agile Coach** (Niels Platz) [7], **Lead Developer** (Kasper Harry Munck) [8], **Scrum Master** (Eva-Maria Bücking-Rasmussen) [9], **Product Owner** (Birgitte Frendrup Jakobsen) [10], and **Enterprise Architect** (Rasmus Witt) [2]. The strength ratings reflect the number of interviews in which the code is evidenced out of five. The Scrum Master removal case recurs across the appendix because it is the single data point that maps onto codes in all three theories and therefore carries the largest share of the main paper’s argumentative weight.

Reading guide. Each code is documented as a **strength rating** and one to three **indicators**. Ratings take one of four values: **Very strong** (5/5 interviews), **Strong** (3–4/5), **Medium** (1–2/5), or **Counter-evidence** (the pattern is **absent** in the first-person voice of every interviewee; the absence is itself a finding and is documented as such). An indicator is a short interpretive sentence followed by a verbatim interview quote and a citation. Codes that illustrate the paper’s central narrative appear with multiple indicators; codes with weaker support appear with one. The Scrum Master removal case is the appendix’s anchor: a single narrative instance that maps onto all three theories.

A. Weinberg's Overstructured Management

a) DEAL — Attribution to Effort or Ability:

Strength: Very strong (5/5 interviews).

Indicator 1. Outcomes are attributed to managerial effort and to the team's growing capability rather than to circumstance. Agile Coach: **"The first part we needed to do was to stop the bleeding. When you talk about Agile, it is nice to write about and read about, and teams are actually the easy ones. Management is the hard part, because they need to do things differently"** [7].

Indicator 2. Decisions are attributed to the developer's ability to make the call, not to external circumstance. Lead Developer: **"It was a quick conversation with the company about accepting the risk, and then pretty quickly I changed the service that handles that validation and took it out of production. Before, I probably would have started looking into other ways or alternatives"** [8].

Indicator 3. Long career trajectories are narrated in terms of accumulated effort and ability. Enterprise Architect: **"I followed problems upward—from developer to tech lead to solution architect to EA. I kept encountering systemic issues that required broader influence to solve"** [2].

b) DEAL — Attribution to Difficulty or Luck:

Strength: Counter-evidence. The pattern is **absent** in the first-person voice of every interviewee.

Indicator 1. The only Difficulty/Luck attribution in the entire data set is **inferred** from a third party. The Agile Coach describes a removed manager who treated the scrum master role as a difficulty to be removed rather than as an ability the teams had not yet built for themselves: **"Then a new manager came in and fired all the Scrum Masters, and things stopped. They created a new application called the [Guba Super Tool], but they did not know what their customers wanted"** [7]. No current Norlys manager in the data attributes failure to task difficulty or to bad luck. The absence is itself a finding: it confirms the main paper's claim that Norlys avoids the Difficulty/Luck side of the DEAL model.

c) Anti-Sequence — Rejection of a Fixed Transformation Order:

Strength: Very strong (5/5 interviews).

Indicator 1. The transformation is framed as evolution rather than revolution. Agile Coach: **"You can take revolutions... or you can take evolutions, which is what we are doing here. Evolutions have less risk because they still allow you to deliver value during the transition. A revolution risks that you die because you did not succeed in your revolution"** [7].

Indicator 2. Practices are knowingly retained as temporary stepping stones even when their architects see them as imperfect. Agile Coach: **"The TPM role is an antipattern of**

huge proportions, but it is a stepping stone toward something better" [7]. Product Owner: **"I'm a technical product manager today and it's not something you can find in any Scrum guide at all. So it's a role we have implemented here in Norlys... I should be out of job in about three years"** [10].

Indicator 3. The QBR planning rhythm is described as a plan, not as a sequence. Agile Coach: **"Q Planning is, from my point of view, a necessity where we are right now. But it is not something I want in the future. When we get further, it needs to be in a different shape and format. So currently, we are not adhering to one fixed model. We are moving"** [7].

d) Anti-Choice — Refusal of Binary Options:

Strength: Very strong (5/5 interviews).

Indicator 1. The bottom-up versus top-down binary is explicitly refused. Agile Coach: **"You need to do all of them. This is a terrible thing, because first you need upper management support. If you do not have that, it will fail. Doing it bottom up will at some point hit a Tayloristic way of thinking... I think you need to work on all different levels"** [7].

Indicator 2. The speed-versus-quality trade-off is treated as a continuum, not a fork. Product Owner: **"I would always say that quality is better than being quick. But it's also kind of a balance, because you shouldn't overdo it. So there's no need to deliver 100% if 95% is enough. So, it's a balance, I think"** [10]. Scrum Master: **"I would always say that quality is better than being quick. But it's also kind of a balance"** [9].

Indicator 3. The decision to deviate from process is treated as a spectrum, not a yes/no call. Product Owner: **"How do you assess when to follow the process and when to deviate from it? Oh, it's a gut feeling. I'm sorry to say, but it's very much a gut feeling"** [10].

e) Anti-Modularization — Roles Treated as More Than Interchangeable Parts:

Strength: Strong (2/5 interviews). The anchor case is concentrated here; the second indicator provides an independent test of the same code.

Indicator 1. Anchor case — Scrum Master removal. Agile Coach: **"Then a new manager came in and fired all the Scrum Masters, and things stopped. They created a new application called the [Guba Super Tool], but they did not know what their customers wanted, what people needed in it, or how to obtain that knowledge... That manager has now been removed, which I am happy about. This is not official wording, but now they are going to have Scrum Masters again. They saw that it simply costs too much to remove the catalyst in the teams"** [7].

Indicator 2. The TPM role is treated as a coach, not as a labelled function. Product Owner: **“It’s very different teams and it’s very different product owners who come with a very different background... some product owners need help in this area and others need help in this area and other product owners have done it so long that they just kind of want to do it themselves”** [10]. The role is **not** swappable; it adapts to the specific product owner it serves.

f) *Anti-Iteration — Awareness That Repeating the Same Action Is Not Progress:*

Strength: Strong (4/5 interviews).

Indicator 1. Ceremonies that stop producing value are removed. Scrum Master: **“Something that is part of the scrum handbook would be review and we don’t have that. In my team we have... I think 18 services and applications. So, it’s not really a product team. So, we don’t have one product that we can present to interested parties. And since a lot of it is back-end... how do you present back-end? You can present the code and you can tell it does this but yeah there isn’t really an interest in it”** [9].

Indicator 2. The “real world has a veto right” framing rejects the repetition of pre-formed plans. Agile Coach: **“Theory is super important, but let it hit the real world, and let the real world have a veto right”** [7].

Indicator 3. Past speed-over-quality choices are explicitly not repeated. Product Owner: **“when you have to do it with speed instead of the correct way, we have what we call spaghetti-code. So if something goes wrong, it’s very hard to clean up in and it becomes legacy in the speed of time... So fortunately that’s better”** [10]. Lead Developer: **“You really do not want to be that early in the lifecycle of something where you pick up all the bugs and teething problems of the early versions”** [8].

g) *Anti-Recursion — Use of Multiple Frames:*

Strength: Medium (3/5 interviews).

Indicator 1. The Agile Coach role is defined as crossing the team/management boundary. Agile Coach: **“Scrum Masters are currently focused on the teams: protecting them and developing them. You also need somebody to work between the lines. That is where the Agile coach works with the different line managers”** [7].

Indicator 2. The Scrum Master’s non-IT background is treated as a recursion-fallacy **antidote**. Scrum Master: **“I have to be the one that says: are we sure this is what we want to do? Because sometimes you just get so excited about a new thing”** [9].

Indicator 3. The Enterprise Architect’s career arc is itself an anti-recursion trajectory. Enterprise Architect: **“I followed problems upward—from developer to tech lead to solution**

architect to EA. I kept encountering systemic issues that required broader influence to solve” [2].

B. *Lyytinen, Rose and Yoo’s Hyper-Learning*

a) *Distributed Gate-Keeping:*

Strength: Very strong (5/5 interviews).

Indicator 1. Multiple roles scan for relevant knowledge. Scrum Master: **“My main role is to remove obstacles for the developers and they don’t always say, oh, this is an issue. Help me with this. So sometimes I just have to listen in and just if I hear something where they’re complaining that this shit doesn’t work, then I can jump in and say, how can I help?”** [9].

Indicator 2. Gate-keeping is institutionalized in the prioritization funnel. Product Owner: **“There is some work happening before the queue planning, which the team is not a part of. So, there’s kind of like a funnel where all the features are put in and then the features that the business thinks are most important, they will go through the funnel and then we get to decide when we can do what within the quarter”** [10].

Indicator 3. Architectural gate-keeping is operationalized as friction. Enterprise Architect: **“Using our Kubernetes platform is easy and self-service, while going off-platform requires much more effort and justification. Guidelines exist for security, observability, and operations. If you deviate, you must prove it’s cheaper, faster, and equally reliable”** [2].

b) *Brokering External Knowledge:*

Strength: Strong (4/5 interviews).

Indicator 1. The Telia/Norlys cultural integration is the most explicit instance. Enterprise Architect: **“For example, when we acquired Telia Denmark, their performance-driven culture mixed with Norlys’ existing culture. Over time, we’ve found a balance, but there were clashes initially”** [2]. Agile Coach: **“You have something stale in how they work and how their business model is going to change, and you need to take that into a world owned by customers. If you keep connecting the two without transforming it, Telia would not make money in the long run”** [7].

Indicator 2. Project-level brokering of third-party APIs. Lead Developer: **“If we have had the task presented to us a month or so in advance, we have had time to look into it. For instance, an integration to a new third party: we can go through the API specifications, go through the implementation details, and get close to an estimate”** [8].

Indicator 3. The TPM role brokers between PO and engineering registers. Product Owner: **“Me, for instance, I don’t have any IT education at all. It’s only my experience in Norlys that has gotten me where I am today”** [10].

c) *Simple Design Rules:*

Strength: Very strong (5/5 interviews).

Indicator 1. The “picture” heuristic for too-large stories. Lead Developer: **“We have a rule that if something is estimated at eight, it is basically a picture, meaning it is too uncertain and not ready. It needs to be broken down into smaller pieces”** [8].

Indicator 2. The 80/20 capacity rule. Product Owner: **“we say, okay, you have 80% of your time to do work from these quarterly meetings where you agree on, and then you have 20% capacity, free capacity to, it can be bugs and it can be new assignments or it can be smaller things that have high value for the business”** [10].

Indicator 3. The Q Planning lock and the platform standardization. Agile Coach: **“We have a candidate list that is extracted from [unclear]. It is locked 14 days before”** [7]. Enterprise Architect: **“We currently support both Java and C#. We define a preferred path for new development while still supporting existing expertise”** [2].

d) *Peer Networks:*

Strength: Strong (4/5 interviews).

Indicator 1. Code review as both quality gate and peer learning. Agile Coach: **“Code reviews are both good and bad. They give value in two ways: the inexperienced person learns, and the experienced person teaches”** [7].

Indicator 2. Knowledge-sharing sessions as a calendar-blocked routine. Scrum Master: **“We do have knowledge sharing sessions. So, we have them in the calendar like once a month or we just say this time is for knowledge sharing but if something pops up like if we worked something more on AI for instance then we would just say okay on Tuesday you will present what you found out”** [9]. Lead Developer: **“When we build a new service, we make sure everyone knows what is going on with it: walk through the basics, maybe walk through some of the code. If someone did something in a new way, we bring it to our knowledge-sharing meetings”** [8].

Indicator 3. The architecture studio as cross-team peer forum. Enterprise Architect: **“Proposals are discussed, refined, and approved in our architecture studio with management support”** [2].

e) *Parallel Ambidexterity:*

Strength: Medium (2/5 interviews). The IP sprint is the central evidence; one interviewee is explicit about deliberately violating the formal structure to enact ambidexterity informally.

Indicator 1. Exploration and exploitation co-exist in the same unit. Agile Coach: **“Right now, we have an event that I am in charge of called Q Planning. We have a candidate list that is extracted from [unclear]. It is locked 14 days before.**

The teams have an innovation and planning sprint. They do their work, then we meet around it, and then we make the plan” [7].

Indicator 2. The Scrum Master breaks the formal structure to enact ambidexterity. Scrum Master: **“With the queue planning setup, the quarter is six sprints, and the sixth sprint is the college... what is it IP standing for innovation and planning so in the process that sprint is meant to be the sprint where you innovate and get smarter and stuff but I cheat. And I think it’s more important that if you find something right now that you need to look into, then do that. Why wait for two and a half months?”** [9].

C. *Mintzberg’s Organization Configurations*

a) *Middle Line — Managerial Layer:*

Strength: Very strong (5/5 interviews). The middle line is the principal site of resistance; it is the empirical anchor of the main paper’s “structural incompleteness” claim.

Indicator 1. Middle management is named as the hardest place to change. Agile Coach: **“Middle management is maybe the hardest place to change. Teams can sometimes be delayed in doing it, because they will live in whatever you give them. I think you need to work on all different levels”** [7].

Indicator 2. The middle line imposes direct supervision on a context that needs catalysts (same anchor case as Anti-Modularization). Agile Coach: **“Then a new manager came in and fired all the Scrum Masters, and things stopped”** [7].

Indicator 3. Leadership turnover at the middle line destabilizes programs. Enterprise Architect: **“Frequent shifts in leadership caused instability and changing priorities”** [2]. Product Owner: **“I would move heaven and earth to get that done. And I sometimes make decisions and then I have to say ‘I’m sorry’ afterwards to a leader of any kind”** [10].

b) *Operating Core — Direct Work:*

Strength: Very strong (5/5 interviews).

Indicator 1. The operating core owns its own quality standard. Agile Coach: **“We create higher standards than what management would demand”** [7].

Indicator 2. Trust in the operating core substitutes for direct supervision. Lead Developer: **“The trust is on the team. There is no micromanagement. It is: come to me if there is a problem. Otherwise you know best what to do”** [8].

Indicator 3. The operating core’s contentment is the Scrum Master’s success metric. Scrum Master: **“I think if it’s a good sprint for me, I think it’s a sprint where the developers are content and have had a good sprint, so if they have good days, it makes me happy. It makes my work easy as well”** [9].

c) *Technostructure — Analyst and Standardization Functions:*

Strength: Very strong (5/5 interviews).

Indicator 1. Finance imposes a standardized output on every story. Agile Coach: **“Recently, I had to present something from finance: a new finance model where they asked that every story, before it is closed, should have story points and an application connected to it, so they could know cost”** [7].

Indicator 2. The prioritization funnel is a technostructure artefact. Product Owner: **“in Norlys you take all the demands from all the business areas and then they have a dogfight somewhere in the leadership and then they come out from one to six”** [10].

Indicator 3. Platform teams standardize the engineering environment. Lead Developer: **“We have platform teams that work on improving the developer experience. They maintain a reference framework project that any team can pull down and use, with all the agreed patterns already baked in: pipelines, deployment into our Kubernetes cluster, everything”** [8].

d) *Support Staff — Internal Services:*

Strength: Strong (3/5 interviews).

Indicator 1. Agile coaches are a support-staff innovation. Agile Coach: **“You also need somebody to work between the lines. That is where the Agile coach works”** [7].

Indicator 2. The TPM role is a support-staff invention. Product Owner: **“I’m a technical product manager today and it’s not something you can find in any Scrum guide at all. So it’s a role we have implemented here in Norlys to help our product owners”** [10].

Indicator 3. Enterprise architecture is support staff in the formal sense. Enterprise Architect: **“Not as a single role. Platform teams, agile processes, and EAs collectively perform that function”** [2].

e) *Strategic Apex — Top Management:*

Strength: Strong (3/5 interviews).

Indicator 1. The IT director enables the transformation. Agile Coach: **“Luckily, we can start from the top. Our IT director, Christian Nord, actually has knowledge of what Agile is”** [7].

Indicator 2. Apex leadership is constrained by the cooperative-ownership structure. Enterprise Architect: **“The representatives can propose and influence decisions. While the CEO and board drive strategy, they must argue their case and get approval from this representative body. Nothing is guaranteed to pass”** [2].

Indicator 3. The apex tries to override the operating core, with mixed success. Product Owner: **“They have tried. One actually succeeded, but fortunately I had some backup, so**

it was with the same success criteria. Alright, give me the value. So they delivered it. Some hasn’t succeeded. It was just a no” [10].

f) *Pull to Lead:*

Strength: Strong (3/5 interviews).

Indicator 1. Apex enables the agile method. Agile Coach: **“Luckily, we can start from the top. Our IT director, Christian Nord, actually has knowledge of what Agile is”** [7].

Indicator 2. Cooperative ownership is the strongest apex pull in the data. Enterprise Architect: **“It stems from the ownership structure—a cooperative model where around 800,000 households are owners. Representatives elected by these owners have final approval authority on major decisions, which strongly influences company culture”** [2].

g) *Pull to Rationalize:*

Strength: Very strong (5/5 interviews).

Indicator 1. Friction is a deliberate design choice. Enterprise Architect: **“Through developer experience and friction. For example, using our Kubernetes platform is easy and self-service, while going off-platform requires much more effort and justification”** [2].

Indicator 2. QBR imposes standardization on planning. Lead Developer: **“With the QBR, it is much clearer: the business has acknowledged that these are the most important things for the next three months, and it gives a much more stable view of the focus areas”** [8].

Indicator 3. Release processes and change gates are imposed from above. Scrum Master: **“In Norlys anyway, there is a change process, so you have to report, and you have to make sure that when we release, we don’t do it in peak hours, for instance”** [9].

h) *Pull to Professionalize:*

Strength: Very strong (5/5 interviews).

Indicator 1. The operating core demands justification from the business. Product Owner: **“now it’s not the business who is in a position to say, ‘You have to do this.’ Now I can say, ‘I would love to do this, but you have to tell me why’”** [10].

Indicator 2. The operating core sets its own quality bar. Agile Coach: **“We create higher standards than what management would demand”** [7].

Indicator 3. Internal experts coordinate the work. Enterprise Architect: **“We have internal experts in billing, commercial products, utilities, etc., and we involve them as needed”** [2].

i) *Pull to Collaborate:*

Strength: Very strong (5/5 interviews).

Indicator 1. Q Planning is the central cross-team liaison device. Agile Coach: **“Right now, we have an event that I am in charge of called Q Planning... The teams have an**

innovation and planning sprint. They do their work, then we meet around it, and then we make the plan” [7].

Indicator 2. QBR formalizes horizontal coordination. Lead Developer: “Then we have conversations with the other teams, because each team has their own backlog. We need to agree on timing: if you need this from us in two weeks, we need to start here. If another team needs to deliver to the self-service portal by August, then they need to finish their part here, and we need to deliver ours here. That way we figure out how everything fits together” [8].

Indicator 3. The architecture studio is a collaboration forum. Enterprise Architect: “Proposals are discussed, refined, and approved in our architecture studio with management support” [2].

j) *Pull to Balkanize:*

Strength: Counter-evidence. The pattern is **resisted** in most of the data and **evidenced** only once, in the Copenhagen acquisition.

Indicator 1. The post-merger integration actively resists balkanization. Agile Coach: “We started out by separating them into two bubbles when we prioritized. Then we found out that the connections became more and more frequent between the parts. That meant finding a uniform way for them to relate to each other needed to make sense” [7].

Indicator 2. One balkanization risk is named. Enterprise Architect: “Yes. There’s a more competitive culture and higher churn in Copenhagen compared to other regions. Big city dynamics play a role” [2]. This is the single explicit evidence for Pull to Balkanize in the data.

D. The Anchor Case (Cross-Cutting)

Strength: A single narrative instance that maps onto codes in all three theories.

The Scrum Master removal case illustrates Weinberg’s Anti-Modularization, Lyytinen’s loss of Distributed Gate-Keeping and a Peer-Network node, and Mintzberg’s middle-line reversal of a Pull to Collaborate. The same passage:

- **Weinberg (Anti-Modularization):** “Then a new manager came in and fired all the Scrum Masters” [7].
- **Lyytinen (DGK / PN loss):** “They created a new application called the [Guba Super Tool], but they did not know what their customers wanted, what people needed in it, or how to obtain that knowledge” [7].
- **Mintzberg (ML reverses P-COL):** “That manager has now been removed, which I am happy about. This is not official wording, but now they are going to have Scrum Masters again. They saw that it simply costs too much to remove the catalyst in the teams” [7].

The case is treated as the anchor of the main paper’s argument because it is the single data point that lets the paper move from parallel application of three theories to sequential application of three theories on the same data point, allowing the logical chain (fallacies → routines → structure) to be walked through in a single narrative.

E. Counter-Evidence and Absences

Following the quality criterion of counter-evidence, the following absences are noted explicitly so that the main paper’s claims do not over-extend the data.

Following the quality criterion of counter-evidence, the following absences are noted explicitly so that the main paper’s claims do not over-extend the data.

- **DEAL-DL is absent in the first-person voice of every interviewee.** No current Norlys manager in the data attributes failure to task difficulty or bad luck. The only trace is the **inferred** attribution of the removed manager in Niels’s account. The main paper’s anti-overstructured claim therefore rests on interviewees’ self-reports—a positive-bias risk that the paper should acknowledge.
- **Parallel Ambidexterity (PA) is the least evidenced Lyytinen code (2/5).** Niels’s IP-sprint description and Eva-Maria’s “I cheat” passage are the only PA evidence. The main paper should not over-claim this routine; the claim should be anchored here, with corroborating rather than universal evidence.
- **Anti-Modularization is the most unevenly evidenced Weinberg code (2/5).** Niels provides the anchor; Birgitte provides a second case (the TPM role). Kasper, Eva-Maria and Rasmus do not address it directly. The paper’s strongest anti-modularization claim is therefore concentrated in two interviewees.
- **Pull to Balkanize is the only Mintzberg code where the paper’s claim is resistance, not presence.** Niels and Birgitte emphasize integration; Rasmus acknowledges the Copenhagen culture’s higher churn. The paper should report both sides.